Robust error-minimization in the genetic code across physicochemical metrics and variant codes: a graph-theoretic analysis in GF(2)^6

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Abstract

The standard genetic code has long been recognized as reducing the impact of point mutations, but the robustness of this property across physicochemical metrics and across naturally variant codes has not been systematically quantified. We embed the 64 codons in $\mathrm{GF}(2)^6$, yielding a 6-dimensional hypercube subgraph of the complete single-nucleotide mutation graph K_4^3, and introduce a weight parameter $\rho \in [0, 1]$ to interpolate between these mutation neighborhoods. Under a block-preserving null ($n = 10\{,\}000$ permutations), the standard code is significantly low-cost under four distinct amino acid distance metrics—Grantham ($p = 0.006$), Miyata ($p < 0.001$), polar requirement ($p = 0.003$), and Kyte–Doolittle hydropathy ($p = 0.001$)—and this signal strengthens as all single-nucleotide substitutions are included ($\rho \to 1$). Across NCBI translation tables, 24 of 25 variant codes retain comparable optimality after Benjamini–Hochberg correction (mean quantile 1.4%).

Natural codon reassignment events show ≈3-fold depletion of topology-breaking moves—changes that fragment an amino acid's codon family—relative to the landscape of possible reassignments (22% observed vs 66–73% possible; RR $= 0.31$–0.34; permutation $p \leq 10^{-4}$), robust to adjacency definition (Q_6 vs K_4^3) and phylogenetic clade exclusion. Discrete-choice (conditional logit) modeling shows that topology avoidance is not explained by physicochemical cost alone: each feature class adds substantial independent explanatory power (ΔAICc ≥ 67; Spearman $r_s = 0.15$ between predictors). Several variant-code lineages with topology-breaking reassignments show elevated tRNA gene counts for the reassigned amino acid (worst-case Stouffer $p = 0.045$, 18 tRNAscan-SE–verified genomes), consistent with compensatory routes to reassignment accommodation. A retrospective reanalysis of eight genome recoding datasets (>248,000 codon-level observations) indicates that codon-family boundary crossing does not predict transcriptomic perturbation, whereas local neighborhood mismatch aligns with established recoding-burden covariates.

Keywords: genetic code evolution; error minimization; graph-theoretic analysis; GF(2)^6 codon reassignment; physicochemical distance; tRNA gene duplication; conditional logit; genome recoding

1 Introduction

The standard genetic code maps 61 sense codons to 20 amino acids through a pattern that has long been recognized as non-random. Woese (1965) first noted that similar codons tend to encode amino acids with similar physicochemical properties, and Freeland and Hurst (1998) demonstrated quantitatively that the standard code sits in approximately the top 10^{-6} percentile of random codes for mutational error minimization under the Woese polar requirement distance. These findings established that the code's structure reduces the fitness impact of point mutations, but the question of whether this optimality is specific to particular physicochemical metrics or represents a robust property across several distinct physicochemical parameterizations has not been systematically addressed.

Every codon comprises three nucleotides drawn from $\{C, U, A, G\}$. By choosing a bijection $\varphi : \{C, U, A, G\} \to \mathrm{GF}(2)^2$—for instance, $C \mapsto (0, 0)$, $U \mapsto (0, 1)$, $A \mapsto (1, 0)$, $G \mapsto (1, 1)$—each codon maps to a vertex of the 6-dimensional binary hypercube $Q_6 = \mathrm{GF}(2)^6$. The genetic code then becomes a *coloring* of Q_6 by 21 labels (20 amino acids plus the stop signal), and single-nucleotide mutations correspond to edges of Q_6 or, more precisely, to a subgraph of the complete mutation graph K_4^3.

This representation is not new in principle—binary encodings of the genetic code appear in the mathematical biology literature (e.g., (Antoneli and Forger, 2011)). Its value lies not in the encoding itself but in the analytical decomposition it enables: the complete single-nucleotide mutation graph K_4^3 (288 edges) separates into 192 Hamming-distance-1 edges (single-bit changes) and 96 within-nucleotide distance-2 edges (both bits of one nucleotide position flip simultaneously), allowing systematic interpolation via a weight parameter $\rho \in [0, 1]$. Note that this decomposition does not correspond to the biological transition/transversion partition: under 16 of the 24 possible 2-bit encodings, the Hamming-1 edges contain an equal mixture of transitions and transversions per nucleotide position (2 of each among 4 Hamming-1 pairs), while the remaining 8 encodings place both transitions on diagonal (Hamming-2) edges. The ρ parameter should therefore be interpreted as a diagonal-edge inclusion weight, not a transition/transversion weight. Previous work has not exploited this decomposition to test error-minimization across multiple physicochemical metrics, nor extended the analysis to variant genetic codes. We address three questions:

1. **Is the standard code optimal under multiple physicochemical metrics?** We test whether the standard code's edge-mismatch score is extreme relative to block-preserving null models across four distinct physicochemical distance metrics (Grantham, Miyata, Woese polar requirement, and Kyte–Doolittle hydropathy), extending Freeland and Hurst (1998) from a single metric to a cross-metric robustness envelope.
2. **Is this structure preserved by evolution?** We ask whether variant genetic codes (NCBI translation tables 2–33) maintain error-minimization, and whether natural codon reassignment events preferentially avoid disrupting the topological connectivity of amino acid codon families.
3. **What are the genomic correlates of disruption?** We test whether organisms whose variant codes break the connectivity of an amino acid's codon graph show elevated tRNA gene copy numbers for the affected amino acid, using tRNAscan-SE–verified data from 10 genomes across 4 eukaryotic supergroups.

In pursuing these questions, we also evaluated several deeper algebraic conjectures from a companion technical note (Clayworth, 2026)—including claims about PSL(2,7) symmetry and holomorphic embeddings—and report their falsification as boundary conditions on the framework's scope (Supplementary Material).

The paper is organized as follows. Section 2 describes the encoding formalism, graph decomposition, null models, and statistical methods, including a new retrospective cross-study reanalysis of eight published genome recoding datasets. Section 3 presents the four supported findings (cross-metric coloring optimality, per-table preservation, ρ-robustness, and topology avoidance), the suggestive tRNA enrichment result, the cross-study reanalysis of synthetic recoding outcomes, and exploratory observations. Section 4 discusses the graph-theoretic interpretation, the relationship to frozen-accident versus adaptive hypotheses (Koonin and Novozhilov, 2009), and a three-layer decomposition of how codon-space structure relates to recoding outcomes. All analyses are reproducible via the open-source `codon-topo` pipeline (version 0.3.1, seed 135325).

2 Methods

2.1 Binary encoding of the genetic code

We encode each nucleotide base as a 2-bit vector via the default bijection $\varphi : C \mapsto (0,0)$, $U \mapsto (0,1)$, $A \mapsto (1,0)$, $G \mapsto (1,1)$. A codon $b_1 b_2 b_3$ is then represented as the concatenation $\varphi(b_1) \parallel \varphi(b_2) \parallel \varphi(b_3) \in \mathrm{GF}(2)^6$. The 64 codons become the 64 vertices of the 6-dimensional hypercube Q_6.

Under this encoding, two codons that differ by a single-nucleotide substitution in which only one bit of the 2-bit pair changes are adjacent in Q_6 (Hamming distance 1). However, transversions that flip both bits within a nucleotide position correspond to Hamming distance 2. The full single-nucleotide mutation graph is therefore K_4^3 (the Cartesian product of three complete graphs on 4 vertices), which contains Q_6 as a subgraph. To address the concern that Q_6 misses approximately one-third of single-nucleotide mutations, we introduce the weighted score F_ρ that interpolates between pure Q_6 ($\rho = 0$) and full K_4^3 ($\rho = 1$); see Section 2.3.

There are 24 distinct bijections from $\{C, U, A, G\}$ to $\mathrm{GF}(2)^2$ (all permutations of $4! = 24$ assignments). All encoding-dependent results are tested across all 24 encodings; encoding-invariant properties (such as Serine's disconnection at $\varepsilon = 1$, where ε denotes the Hamming distance threshold in $\mathrm{GF}(2)^6$) are noted as such. Coloring optimality is significant ($p < 0.05$) under all 24 encodings; full encoding-sensitivity results are provided in the Supplementary Material.

2.2 Edge-mismatch objective function

The genetic code assigns each vertex $v \in Q_6$ a label $c(v) \in \mathcal{A}$, where $\mathcal{A}$ comprises the 20 amino acids and the stop signal. The edge-mismatch score is

$$F(c) = \sum_{\{v,w\}:d(v,w)=1} \Delta(c(v), c(w)) \tag{1}$$

where the sum ranges over all 192 edges of Q_6 (pairs of vertices at Hamming distance 1), and Δ is a physicochemical distance between amino acids. We test four distinct distance metrics: the Grantham (1974) composite distance (composition, polarity, volume; range 5–215), the Miyata et al. (1979) normalized Euclidean distance (polarity and volume only; range 0.06–5.13), the Woese polar requirement absolute difference (range 0–8.2; (Woese, 1973, Woese et al., 1966); used by Freeland and Hurst (1998)), and the Kyte and Doolittle (1982) hydropathy absolute difference (range 0–9.0; used by Haig and Hurst (1991)). Synonymous edges contribute 0; edges involving a stop codon receive a fixed penalty scaled proportionally to each metric's maximum. A lower F indicates a more error-minimizing code.

Stop codons are held fixed across all null models (Section 2.3), so the stop-codon contribution is a constant offset. Sensitivity analysis across penalty values (0, 150, 215, 300) confirms this is immaterial to the ranking.

2.3 Null models

2.3.1 Block-preserving null (Freeland–Hurst)

Following Freeland and Hurst (1998), we group the 64 codons into 16 blocks of 4, defined by shared first-two-base prefix. Each block's internal pattern of amino acid assignments is preserved, but the mapping of patterns to blocks is permuted uniformly at random. Blocks containing stop codons are held fixed. This null preserves the synonymous codon contiguity (wobble degeneracy) inherent in the genetic code, providing a stringent test: the observed code must beat random codes that share its degeneracy architecture.

For the standard code, we drew $n = 10,000$ null samples (seed 135325) and computed the conservative p-value as $\frac{k+1}{n+1}$, where k is the number of null scores below the observed F.

2.3.2 Per-table null

Each of the 25 NCBI translation tables was tested against its own block-preserving null ($n = 1\{,\}000$ per table, seeds = base seed + table ID). P-values were corrected for multiple comparisons using the Benjamini and Hochberg (1995) false discovery rate (FDR) procedure.

2.3.3 Graph family G_ρ and rho sweep

We define a family of mutation graphs G_ρ parameterized by $\rho \in [0, 1]$, where $G_0 = Q_6$ (192 Hamming-1 edges) and $G_1 = K_4^3$ (all 288 single-nucleotide substitution edges). The weighted mismatch score on G_ρ is

$$F_{\rho(c)} = F_{\text{Hamming-1}}(c) + \rho \cdot F_{\text{diagonal}}(c) \tag{2}$$

where F_{diagonal} sums over the 96 within-nucleotide distance-2 edges. This formulation treats Q_6 not as the biological object but as one endpoint of a continuous interpolation toward the full mutation graph. We evaluated 5 values of ρ ($n = 1\{,\}000$ null samples per ρ, common-seed design).

2.3.4 Table-preserving permutation null (topology avoidance)

To test whether natural codon reassignment events avoid disrupting amino acid connectivity, we enumerated all 1,280 possible single-codon reassignments from the standard code. For each amino acid a, let G_a^1 denote the subgraph induced on codons assigned to a with edges between codons at binary Hamming distance ≤ 1. A reassignment is *topology-breaking* if it increases the number of connected components of G_a^1 for any amino acid a. We define a de-duplicated reassignment event as a unique (codon, target amino acid) pair observed across the 25 NCBI translation tables, yielding 27 events from 61 total table-specific changes. We then compared the observed rate of topology-breaking among 27 de-duplicated natural reassignment events to the null expectation.

Two tests were applied: (i) a hypergeometric test treating the observed events as a sample from the finite landscape of possible reassignments ($N = 1\{,\}280$, $K = 931$ topology-breaking under Q_6, $n = 27$ observed, $x = 6$ topology-breaking observed); and (ii) a table-preserving permutation test ($n = 10\{,\}000$, seed 135325) that permutes target amino acids among a table's reassigned codons, preserving within-table codon and target structure to address phylogenetic non-independence. Risk

ratios with 95% log-normal confidence intervals were computed as $\mathrm{RR} = (x/n)/(K/N)$. To test robustness to the adjacency definition, both tests were repeated under K_4^3 adjacency ($K = 846$ topology-breaking), where codon-family connectivity is defined by nucleotide-level single-substitution adjacency (encoding-independent).

2.3.5 Conditional logit model of reassignment choice

To test whether topology avoidance contributes independent explanatory power beyond physicochemical optimization, we fit event-level conditional logit (discrete-choice) models to natural reassignment events. Each observed reassignment is treated as a choice from the set of all $\approx 1\{,\}280$ possible single-codon reassignments available at the current code state. For a table with k reassignment events, the code state evolves sequentially: at each step, the model assigns a probability to each candidate move based on a linear utility score, and the observed move is compared against all alternatives.

For each candidate move m from code state C, we computed three features: (i) Δ_{phys}, the change in local Grantham mismatch cost summed over Hamming-1 edges incident to the reassigned codon; (ii) Δ_{topo}, the total increase in connected components across all amino acid codon graphs at $\varepsilon = 1$; and (iii) Δ_{tRNA}, the Hamming distance from the reassigned codon to the nearest codon already encoding the target amino acid (a heuristic proxy for tRNA repertoire disruption). The conditional logit probability of observing move m^* is

$$P(m^* \mid \mathcal{N}(C)) = \frac{\exp(\boldsymbol{w}^\top \boldsymbol{x}_{m^*})}{\sum_{m \in \mathcal{N}(C)} \exp(\boldsymbol{w}^\top \boldsymbol{x}_m)} \tag{3}$$

where $\boldsymbol{w}$ is the weight vector and $\boldsymbol{x}_m$ is the feature vector for move m. Features were z-scored across all candidates for numerical stability.

Since the temporal ordering of reassignment events within a table is unknown, we marginalized the likelihood over all $k!$ orderings for tables with $k > 1$ changes:

$$L_{\mathrm{table}} = \frac{1}{k!} \sum_{\sigma \in S_k} \prod_{s=1}^{k} P\big(m^*_{\sigma(s)} \mid \mathcal{N}(C_{\sigma,s})\big) \tag{4}$$

For the largest table (yeast mitochondrial, $k = 6$), we averaged over 24 orderings (a random subset of the 720 possible). The total likelihood is the product across all 23 tables with reassignment events (61 total event-steps).

Four nested models were compared: M1 (physicochemistry only, $w_{\mathrm{topo}} = w_{\mathrm{tRNA}} = 0$), M2 (topology only, $w_{\mathrm{phys}} = w_{\mathrm{tRNA}} = 0$), M3 (physicochemistry + topology, $w_{\mathrm{tRNA}} = 0$), and M4 (all three features). Weights were estimated by maximum likelihood (scipy.optimize, Nelder–Mead with L-BFGS-B refinement). Model comparison used the corrected Akaike Information Criterion (AICc) and likelihood-ratio tests for nested pairs.

2.3.6 Fisher–Stouffer test for tRNA enrichment

For each variant-code organism paired with a phylogenetically proximate standard-code control, we built a 2×2 contingency table comparing the proportion of tRNA genes encoding the reassigned amino acid versus all other amino acids. The sampling model treats tRNA gene counts as draws from a hypergeometric distribution conditional on the row and column marginals (total tRNAs per organism, total tRNAs for the focal amino acid), which is appropriate when the question is whether a specific amino acid's share of the tRNA repertoire differs between two genomes. Fisher's exact test (one-sided, alternative = "greater") was applied per pairing, and p-values were combined via Stouffer's Z method.

To address non-independence from shared control organisms, we constructed a conflict graph (edges connect pairings sharing an organism) and enumerated all maximal independent sets (MIS) via the Bron–Kerbosch algorithm with pivoting. For each MIS with ≥ 2 members, we computed the Stouffer combined p-value and report the worst-case (maximum p) across all MIS as the primary conservative test.

tRNA gene counts for 17 organisms were verified by running tRNAscan-SE 2.0.12 (Chan and Lowe, 2019) with Infernal 1.1.4 on NCBI genome assemblies (eukaryotic mode for ciliates, bacterial mode for *Mycoplasma*). Counts for 3 additional organisms were drawn from published literature. The dataset comprises 24 pairings across 5 variant genetic codes (NCBI translation tables 4, 6, 10, 15, and 31) and 3 standard-code controls.

2.4 Retrospective cross-study reanalysis of synthetic recoding outcomes

To test whether GF(2)^6 topology predicts codon recoding outcomes in synthetic biology experiments, we performed a retrospective cross-study reanalysis across eight published genome recoding datasets. For each codon substitution reported in these studies, we computed three topology features under the default encoding ($C \mapsto (0,0)$, $U \mapsto (0,1)$, $A \mapsto (1,0)$, $G \mapsto (1,1)$):

1. **Boundary crossing** ($\varepsilon = 1$): whether the source and target codons lie in different connected components of the amino acid's codon graph at Hamming distance 1. Generalized beyond serine to all amino acids with potential disconnections (including leucine, arginine).
2. **Local mismatch change** (ΔF_{local}): the difference in total Grantham distance summed over all Hamming-1 neighbors between the target and source codon positions: $\Delta F_{\text{local}} = \sum_{n \in N(t)} \Delta(c(t), c(n)) - \sum_{n \in N(s)} \Delta(c(s), c(n))$, where $N(v)$ denotes the 6 Hamming-1 neighbors of vertex v in Q_6.
3. **Hamming distance**: the number of bit positions differing between source and target vectors in GF(2)^6.

Codon positions were extracted from published GenBank genome files by CDS-level comparison using Biopython (Cock et al., 2009), with codon frame offsets (`/codon_start`) handled explicitly. CDS features were matched between parent and recoded genomes by locus tag. The primary analysis used the Syn57 dataset (Robertson et al., 2025), which provides a within-study contrast: 37,146 serine recodings cross the UCN↔AGY family boundary, while 22,859 alanine recodings remain within the GCN family. Syn57 RNA-seq differential expression data (Data file S10) were used as the outcome measure. Design-to-final genome deviations were identified by comparing the vS33A7 design genome (Data file S2) against the verified Syn57 genome (Data file S8), filtering genes with >10 CDS-level differences as structural rearrangements (sensitivity analysis across thresholds 3–50 reported in Supplementary Table SX). The Ostrov et al. (2016) segment viability data (Table S4) were analyzed as a case-control comparing segments with and without lethal design exceptions; Bonferroni correction was applied for three simultaneous tests.

All cross-study reanalysis code is available in the `codon_topo.analysis.codonsafe` subpackage (version 0.3.1) and can be run via `codon-topo codonsafe`. Raw data files and download provenance are documented in `data/codonsafe/DATA_MANIFEST.md`.

2.5 Claim hierarchy

All 15 scientific claims evaluated in this work are registered in a formal claim hierarchy with pre-specified status categories (Table 1): SUPPORTED (passes rigorous null, $p < 0.01$), SUGGESTIVE

Status	Claim	Null model	p-value
Supported	Cross-metric coloring optimality	FH block-preserving × 4 metrics	≤ 0.006
	Per-table optimality (24/25)	FH per-table + BH-FDR	< 0.05
	ρ-robustness ($\rho \in [0, 1]$)	FH weighted edges	0.006
	Topology avoidance (Q_6 + K_4^3)	Table-pres. perm. + clade excl.	10^{-4}
Suggestive	tRNA enrichment (MIS worst-case)	Fisher–Stouffer (MIS)	0.045
Exploratory	Bit-position bias (dedup.)	χ^2 uniform	0.075
	Mechanism boundary conditions	Descriptive	—
	Atchley F3/Serine convergence	Qualitative	—
	Disconnection catalogue (4 cases)	Descriptive	—
Falsified	KRAS–Fano clinical prediction	Fisher–Bonferroni	1.0
Rejected	Serine min-distance-4 invariant	Counterexample	—
	PSL(2,7) symmetry	Literature	—
	Holomorphic embedding	Verification	—
Tautological	Two-fold bit-5 filtration	—	—
	Four-fold prefix filtration	—	—

Table 1: Summary of 15 evaluated claims. "FH" = Freeland–Hurst. "MIS" = maximal independent set enumeration. p-values are the most conservative test per claim. Shaded rows: claims that failed validation.

(trend-level, $p < 0.05$), EXPLORATORY (hypothesis-generating), REJECTED (falsified or pre-rejected in literature), FALSIFIED (directly contradicted by data), and TAUTOLOGICAL (true by construction). The full hierarchy with justifications is available in the supplementary materials.

3 Results

3.1 Cross-metric coloring optimality

Under the Freeland–Hurst block-preserving null with $n = 10{,}000$ permutations, the standard genetic code is significantly low-cost across all four physicochemical distance metrics examined (Table 2). For Grantham distance, the observed score is $F = 13{,}477$ versus a null mean of $14{,}954 \pm 628$ ($z = 2.35$, quantile 0.6%, $p = 0.006$). The same pattern holds for Miyata distance ($F = 235.3$ vs 275.7 ± 12.9, $z = 3.14$, $p < 0.001$), Woese polar requirement ($F = 347.3$ vs 410.5 ± 23.1, $z = 2.73$, $p = 0.003$), and Kyte–Doolittle hydropathy ($F = 457.2$ vs 570.7 ± 40.9, $z = 2.78$, $p = 0.001$). Relative to null expectations, the observed code lowers the mismatch score by 9.9%, 14.6%, 15.4%, and 19.9%, respectively. Beta-posterior credible intervals for the empirical p-values remain entirely below 0.01 for all four metrics (Supplementary Table S1). Since Freeland and Hurst (1998) established optimality using polar requirement alone, the cross-metric concordance demonstrates that error-minimization is robust across multiple distinct physicochemical parameterizations, not an artifact of any single distance metric. Under a weaker degeneracy-only null that preserves only codon family sizes without maintaining block contiguity, the signal strengthens substantially ($z > 9$, $p < 10^{-4}$ for all metrics), confirming that the block-preserving null provides a stringent test.

Metric	Observed	Null mean ± SD	z	p	Improv.
Grantham (Grantham, 1974)	13,477	14,954 ± 628	2.35	0.006	9.9%
Miyata (Miyata et al., 1979)	235.3	275.7 ± 12.9	3.14	< 0.001	14.6%
Polar requirement (Woese, 1973)	347.3	410.5 ± 23.1	2.73	0.003	15.4%
Kyte–Doolittle (Kyte and Doolittle, 1982)	457.2	570.7 ± 40.9	2.78	0.001	19.9%

Table 2: Cross-metric sensitivity analysis. Block-preserving null ($n = 10{,}000$, seed 135325) under four distinct physicochemical distance metrics. All metrics show significant optimality ($p < 0.01$). Effect size $z = (\mu_{\text{null}} - F_{\text{obs}})/\sigma_{\text{null}}$. "Improv." = percent improvement of observed score over null mean. Under a weaker degeneracy-only null (preserving only codon family sizes, not block structure), all four metrics yield $z > 9$ and $p < 10^{-4}$, confirming the result is not an artifact of the Freeland–Hurst (Freeland and Hurst, 1998) block-preserving design.

Decomposing F by nucleotide position reveals that the second codon position contributes 49.3% of the total mismatch, followed by the first position at 38.2% and the wobble position at 12.5%. This gradient reflects the well-known biochemical hierarchy: second-position mutations are the most physicochemically disruptive, first-position mutations are intermediate, and wobble-position mutations are largely synonymous.

3.2 Robustness across transversion weights

The weighted score F_ρ (Equation 2) remains significantly optimal across all values of ρ from 0 to 1 (Table 3). At $\rho = 0$ (pure Q_6), $p = 0.003$ ($z = 2.48$); at $\rho = 1$ (full K_4^3, all 288 single-nucleotide edges equally weighted), $p < 0.001$ ($z = 3.46$). No value of ρ yields $p > 0.003$. The optimality signal strengthens monotonically as diagonal edges are included, addressing the concern that the Q_6 representation ignores approximately one-third of single-nucleotide mutations.

3.3 Per-table optimality preservation

Of the 25 NCBI translation tables, 24 remain in the top 5% of their own block-preserving null after Benjamini–Hochberg FDR correction (Figure 2, panel C). The mean quantile across all tables is 1.4%. Only translation table 3 (yeast mitochondrial code, 6 codon reassignments) marginally exceeds the 5% threshold.

This result demonstrates that codon reassignment events, despite altering the mapping between codons and amino acids, generally preserve the error-minimization structure of the code. Even the most extensively reassigned code retains near-optimal physicochemical placement.

3.4 Topology avoidance in natural codon reassignments

Among all 1,280 possible single-codon reassignments from the standard code, 931 (72.7%) would create a new amino acid disconnection at Hamming distance $\varepsilon = 1$ in Q_6. By contrast, only 6 of 27 observed natural reassignment events (22.2%) are topology-breaking—a 3.3-fold depletion (hypergeometric $p = 0.00000004776789702630534$; table-preserving permutation $p \leq 10^{-4}$).

This depletion is robust to the choice of adjacency graph. Under the fuller K_4^3 single-nucleotide mutation graph, where codon-family connectivity is defined by nucleotide-level adjacency (encoding-independent), 846 of 1,280 possible reassignments (66.1%) are topology-breaking—somewhat fewer

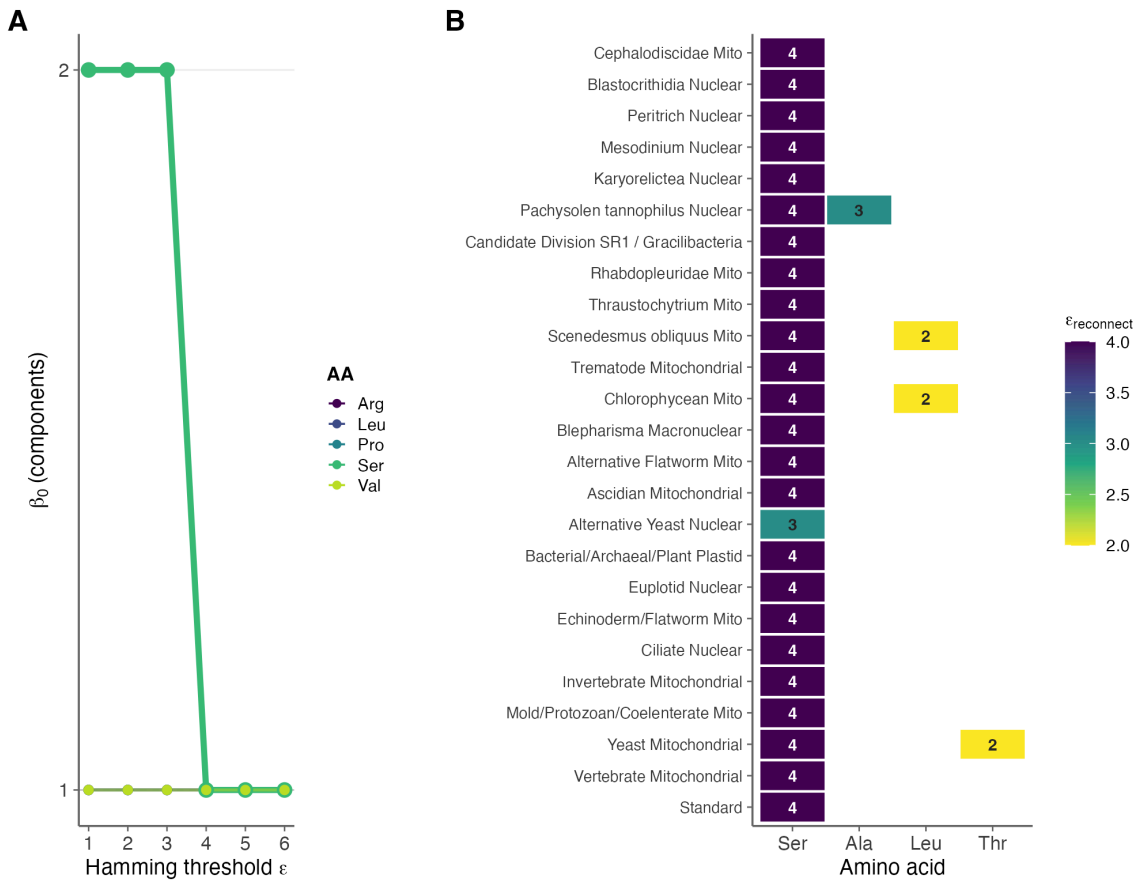

Figure 1: Core topology of the genetic code in $\mathrm{GF}(2)^6$. **(A)** Persistent homology: connected components (β_0) of amino acid codon graphs as a function of Hamming distance threshold ε. Serine (bold) is the only amino acid disconnected at $\varepsilon = 1$ across all 25 NCBI translation tables, reconnecting at $\varepsilon = 4$. **(B)** Disconnection catalogue across all translation tables. Each tile denotes an amino acid disconnected at $\varepsilon = 1$; the number is the reconnection ε. Serine is universally disconnected; Leu, Thr, and Ala appear as variant-code disconnections.

than under Q_6, because some Q_6-disconnected pairs become connected when within-nucleotide Hamming-2 edges are admitted. The observed proportion remains 22.2% (6/27), yielding a 3-fold depletion (RR $= 0.34$, 95% CI $[0.17, 0.68]$; hypergeometric $p = 0.0000030710765502781 02$; permutation $p \leq 10^{-4}$). The avoidance of topology-disrupting reassignment trajectories is therefore not specific to the hypercube subgraph.

To address phylogenetic non-independence (Sengupta et al., 2007), we performed clade-exclusion sensitivity analysis, iteratively removing each major taxonomic group (all ciliates, all metazoan mitochondria, all CUG-clade yeasts, etc.) and retesting. The depletion remains highly significant in every case ($p < 10^{-5}$), confirming that the result is not driven by any single clade.

3.5 Explanatory modeling: topology as an independent predictor

To test whether topology avoidance is reducible to physicochemical optimization, we fit event-level conditional logit models to the 61 reassignment event-steps across 23 variant-code tables, treating each observed reassignment as a choice among $\approx 1\{,\}280$ candidate single-codon moves (Section 2.3.5). The combined physicochemical-plus-topology model (M3) was strongly favored over all alternatives

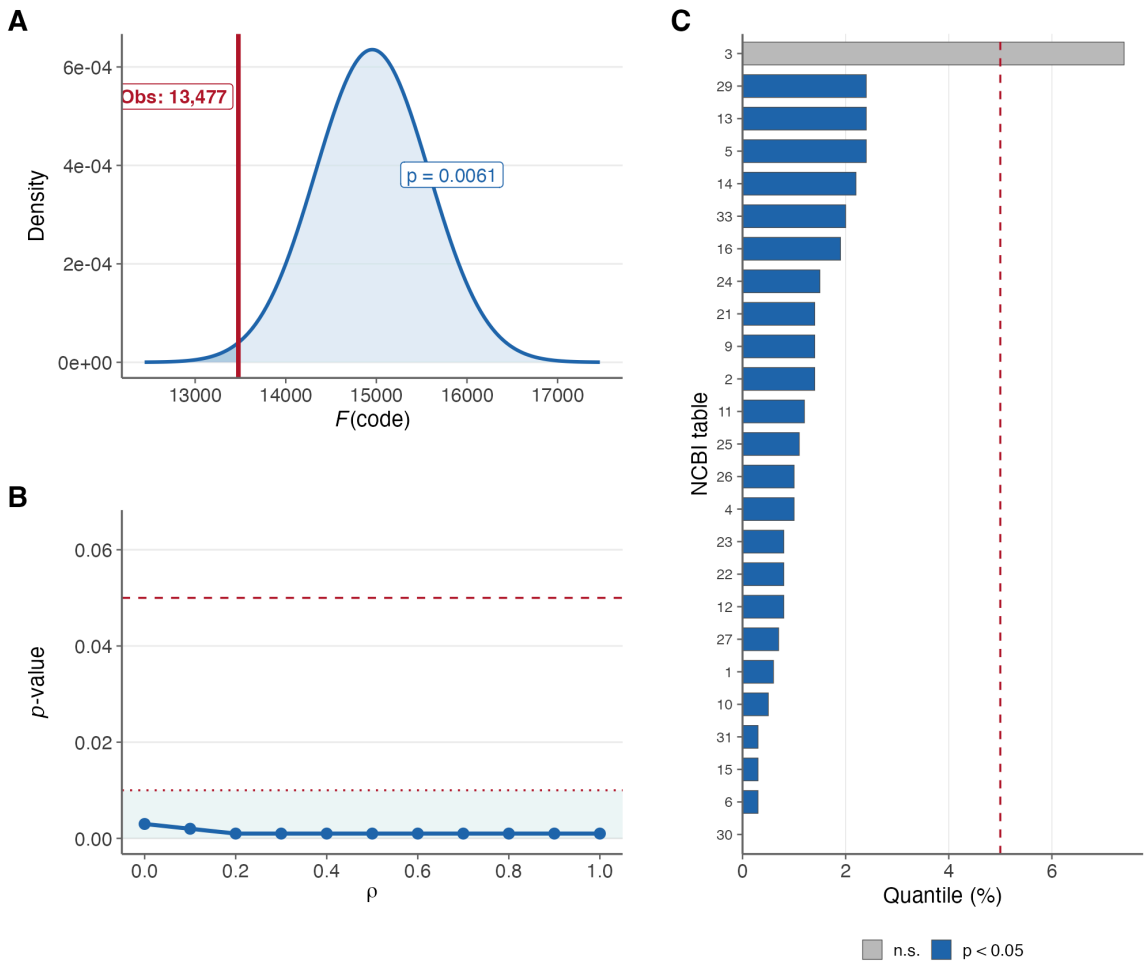

Figure 2: Coloring optimality of the genetic code. **(A)** Null distribution of Grantham edge-mismatch scores F under the Freeland–Hurst block-preserving null ($n = 10{,}000$). The observed standard code (red line, $F = 13{,}477$) falls at the 0.6th percentile ($p = 0.006$). **(B)** p-value across transversion weight ρ from 0 (Q_6) to 1 (full K_4^3); all values below 0.01, with optimality strengthening monotonically. **(C)** Per-table quantile for each of 25 NCBI translation tables under their own block-preserving null ($n = 1\{,\}000$); only table 3 (yeast mito.) exceeds the 5% threshold.

by AICc (AICc $= 682.5$), outperforming the physicochemistry-only model (M1; ΔAICc $= 101.5$) and

ρ	Observed	Null mean	Quantile	*p*-value	*z*
0	13,477	14,998	0.2%	0.003	2.48
0.25	15,335	17,063	0%	< 0.001	2.9
0.5	17,193	19,127	0%	< 0.001	3.22
0.75	19,050	21,192	0%	< 0.001	3.4
1	20,908	23,256	0%	< 0.001	3.46

Table 3: Robustness of coloring optimality across diagonal-edge weight ρ. Block-preserving null, $n = 10{,}000$ per ρ. All p-values < 0.01. Quantiles reported as empirical rank; values $\leq 0.02\%$ indicate ≤ 2 null samples below the observed score.

Metric	Observed	Possible (Q_6)	Possible (K_4^3)	Expected (chance)
Topology-breaking	6 (22.2%)	931 (72.7%)	846 (66.1%)	—
Topology-preserving	21 (77.8%)	349 (27.3%)	434 (33.9%)	—
Total events	27	1,280	1,280	—
Depletion fold	—	3.3×	3×	—
RR (95% CI)	—	—	0.34 (0.17–0.68)	—
Hypergeometric p	0.0000004776789702630534	0.0000307107655027810 2		—
Permutation p	—	$\leq 10^{-4}$	$\leq 10^{-4}$	—

Table 4: Topology avoidance under two adjacency definitions. Q_6: Hamming-1 edges in $\mathrm{GF}(2)^6$; K_4^3: full single-nucleotide mutation graph (encoding-independent). Topology-breaking = creates a new amino acid graph disconnection. RR = risk ratio with 95% log-normal CI. Both adjacency definitions yield highly significant depletion of topology-breaking reassignments.

the topology-only model (M2; $\Delta\mathrm{AICc} = 67$). Adding a heuristic tRNA-complexity proxy did not improve fit (M3→M4: $\Delta\mathrm{AICc} = 1.7$).

Likelihood-ratio tests confirmed that each feature class adds substantial explanatory value to the other: adding topology to physicochemistry yields LR $= 103.7$ ($p \ll 10^{-10}$), and adding physicochemistry to topology yields LR $= 69.1$ ($p \ll 10^{-10}$). The two feature classes are only weakly associated across the full candidate landscape, indicating limited confounding.

In the best-fitting model (M3), evolution favored moves that reduce local physicochemical mismatch ($\hat{\beta}_{\mathrm{phys}} = -0.004$ per Grantham unit) and strongly disfavored moves that increase amino acid codon-family disconnection ($\hat{\beta}_{\mathrm{topo}} = -3.26$ per additional connected component). Under M3, observed natural reassignments rank on average at the 89.5th percentile among all candidate moves (Figure 4, panel B), with the recurrent UGA→Trp reassignment consistently ranking above the 98th percentile. The one notable outlier is the yeast mitochondrial CUU→Thr reassignment (30th percentile), consistent with translation table 3 being the sole marginal exception in the per-table optimality analysis (Section 3.3).

3.6 tRNA enrichment for reassigned amino acids

Organisms with variant genetic codes show elevated tRNA gene copy numbers for the reassigned amino acid relative to standard-code controls. Across 24 pairings (18 variant-code organisms, 6 standard-code controls, spanning 5 variant genetic codes and 18 tRNAscan-SE–verified genome assemblies), Fisher's exact test combined via Stouffer's Z method yields $p = 1.7 \times 10^{-7}$ ($Z = 5.10$). To address non-independence from shared controls, we enumerated all maximal independent sets (MIS) from the conflict graph via Bron–Kerbosch; both MIS (each of size 6) are significant at $p < 0.05$ (worst-case $p = 0.045$).

The most striking case is *Tetrahymena thermophila* (NCBI translation table 6, UAA/UAG reassigned to Gln; (Hanyu et al., 1986)), which carries 54 glutamine tRNA genes—including 39 suppressor tRNAs reading the reassigned stop codons—compared to 3 Gln tRNAs in the standard-code ciliate *Ichthyophthirius multifiliis*, 11 in *Stentor coeruleus* and 3 in *Fabrea salina* (Zhang et al., 2022). The pattern extends across reassignment types. Among Gln-reassignment ciliates, *Pseudocohnilembus persalinus*

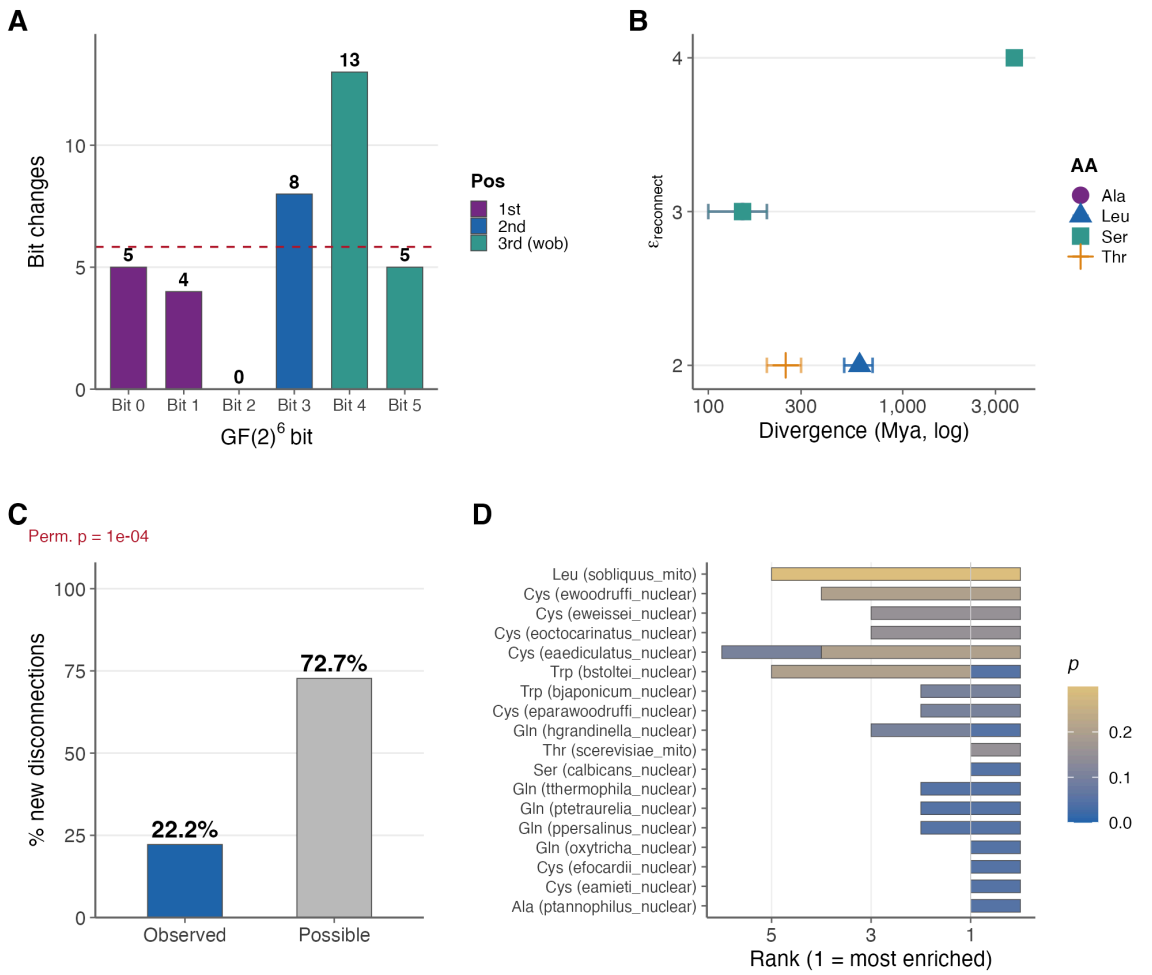

Figure 3: Evolutionary evidence for structural constraints on codon reassignment. **(A)** Bit-position bias: distribution of bit-flips across $\mathrm{GF}(2)^6$ coordinates in natural reassignment events; dashed line = uniform expectation. **(B)** Evolutionary depth calibration: reconnection ε vs estimated divergence age (log scale) for 4 variant-code amino acids. **(C)** Topology avoidance: observed natural reassignments create new disconnections at 22.2% vs 72.7% among all possible single-codon changes (permutation $p \leq 10^{-4}$). **(D)** tRNA enrichment: rank of the reassigned amino acid among all 20 AAs by tRNA gene proportion in variant-code vs standard-code organism pairings; rank 1 = most enriched.

(20 Gln tRNAs including 15 suppressors) and *Halteria grandinella* (Zheng et al., 2021) (9 Gln tRNAs including 3 suppressors) represent independent lineages within Oligohymenophorea and Spirotrichea respectively. Among Cys-reassignment ciliates (translation table 10, UGA→Cys), six tRNAscan-SE–

Model	Description	k	Log $\mathcal{L}$	AICc	ΔAICc
M3	Physicochemistry + topology	2	−339.1	**682.5**	**0**
M4	Physicochemistry + topology + tRNA	3	−338.9	684.2	1.7
M2	Topology only	1	−373.7	749.5	67
M1	Physicochemistry only	1	−391	784	101.5

Table 5: Event-level conditional logit model comparison. Each model assigns a probability to the observed reassignment among $\approx 1\{,\}280$ candidate single-codon moves, with likelihood marginalized over event orderings within each table. k = number of estimated parameters. Bold = best model by AICc.

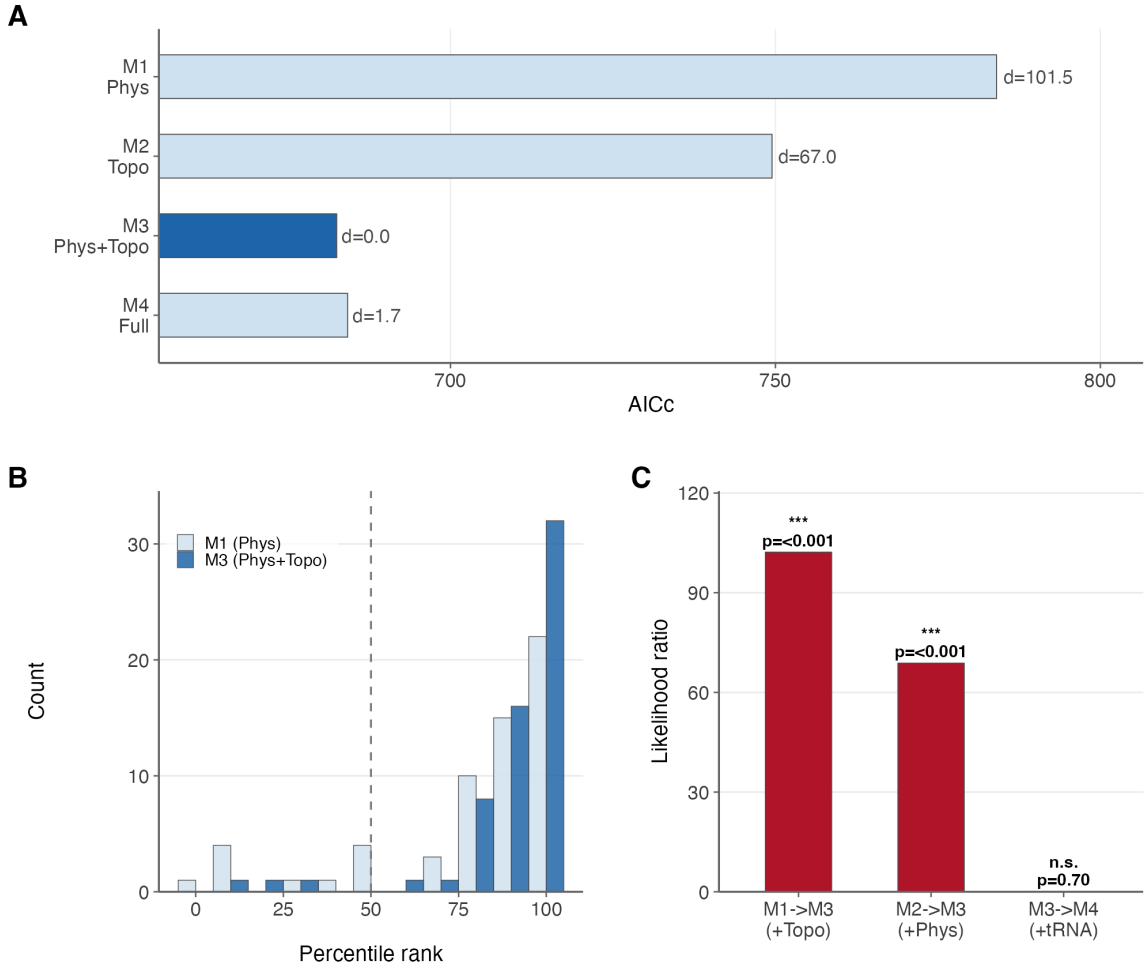

Figure 4: Event-level explanatory modeling of natural codon reassignments. **(A)** AICc comparison of four nested conditional logit models (M1–M4); lower is better. The combined physicochemistry-plus-topology model (M3) is decisively favored (ΔAICc ≥ 67). **(B)** Distribution of observed move percentile ranks under M1 (physicochemistry only, light) versus M3 (combined, dark); M3 concentrates ranks toward the top of the candidate set. Dashed line = chance expectation (50th percentile). **(C)** Likelihood-ratio tests for each feature class added to its complement; both topology and physicochemistry contribute highly significant independent information ($p < 0.001$); the heuristic tRNA proxy does not ($p = 0.70$).

verified *Euplotes* species all carry tRNA-Cys genes with the non-canonical TCA anticodon (reading UGA), with 1–4 such genes per species alongside standard GCA-anticodon Cys tRNAs.

However, the pattern is not universal. *Blastocrithidia nonstop* (translation table 31) reassigned all three stop codons (as did *Condylostoma magnum*, where stop-codon function is context-dependent; (Heaphy et al., 2016)) but achieved UGA→Trp via anticodon stem shortening (5 bp → 4 bp) of tRNA-Trp(CCA), combined with an eRF1 Ser74Gly mutation, rather than gene duplication (Kachale et al., 2023). Similarly, *Mycoplasma* species with UGA→Trp use a single tRNA-Trp with anticodon modification. These boundary cases define a three-tier mechanistic landscape: (i) tRNA gene duplication in large nuclear genomes, (ii) anticodon structural modification in streamlined genomes, and (iii) anticodon base modification in minimal genomes.

Test	n	Statistic	p-value
Fisher+Stouffer (all pairings)	24	$Z = 5.10$	1.7×10^{-7}
Fisher+Stouffer (6 indep.)	6	$Z = 2.13$	0.017
MIS worst-case	**6**	$\boldsymbol{Z = 1.70}$	**0.045**
MIS best-case	6	$Z = 1.71$	0.044

Table 6: Summary of tRNA enrichment tests across 24 pairings (18 tRNAscan-SE–verified genomes, 5 variant genetic codes). MIS = maximal independent set, enumerated via Bron–Kerbosch algorithm on the conflict graph. Both MIS are significant at $p < 0.05$, eliminating the concern that greedy selection biased the independent-pairings result. The MIS worst-case ($p = 0.045$) serves as a robustness bound demonstrating that the result is not driven by a cherry-picked independent subset.

3.7 Score decomposition by nucleotide position

The decomposition of F by nucleotide position (Figure 2, panel b) reveals that the second codon position dominates the error-minimization signal, contributing 49.3% of the total physicochemical mismatch. This aligns with the biochemical observation that second-position mutations cause the largest changes in amino acid hydrophobicity (Freeland and Hurst, 1998, Woese, 1965). The wobble position contributes only 12.5%, consistent with its largely synonymous character under the genetic code's degeneracy structure.

Organism	Code	Tbl. #	Assembly	tRNAs	Reassigned AA
T. thermophila	Variant	6	GCF_000189635.1	718	54 Gln
P. tetraurelia	Variant	6	GCF_000165425.1	216	18 Gln
O. trifallax	Variant	6	GCA_000295675.1	94	8 Gln
P. persalinus	Variant	6	GCA_001447515.1	262	20 Gln
H. grandinella	Variant	6	GCA_006369765.1	130	9 Gln
E. aediculatus	Variant	10	GCA_030463445.1	80	4 Cys
E. focardii	Variant	10	GCA_001880345.2	62	3 Cys
E. parawoodruffi	Variant	10	GCA_021440025.1	149	9 Cys
B. stoltei	Variant	15	GCA_965603825.1	169	6 Trp
B. nonstop P57	Variant	31	GCA_028554745.1	68	2 Trp[*]
M. genitalium	Variant	4	GCA_000027325.1	36	1 Trp[†]
S. coeruleus	Standard	1	GCA_001970955.1	272	—
I. multifiliis	Standard	1	GCF_000220395.1	150	—
F. salina	Standard	1	GCA_022984795.1	89	—

Table 7: Representative organisms verified by tRNAscan-SE 2.0.12 (18 total; 3 additional *Euplotes* and 1 *Mycoplasma* species in supplement). "Tbl. #" = NCBI translation table number. "Reassigned AA" = effective tRNA gene count including suppressor tRNAs. [*]Anticodon stem shortening. [†]Post-transcriptional modification (single tRNA reads UGG+UGA).

3.8 Exploratory observations

3.8.1 Bit-position bias in codon reassignments

The distribution of bit-flips across the 6 coordinates of $\mathrm{GF}(2)^6$ in natural codon reassignments shows apparent positional skew under a uniform null ($\chi^2 = 16.26$, $p = 0.006$, df $= 5$). However, this signal is substantially attenuated after de-duplication to 20 unique (codon, target amino acid) pairs ($p = 0.075$) and vanishes entirely under a codon-preserving permutation null ($p = 1.0$). The apparent bias is therefore explained by which codons are "hot" for reassignment, not by a genuine positional preference in $\mathrm{GF}(2)^6$.

3.8.2 Variant-code disconnection catalogue

A systematic survey across all 25 NCBI translation tables identifies 4 variant-code amino acids with disconnected Hamming graphs at $\varepsilon = 1$: threonine in the yeast mitochondrial code (translation table 3), leucine in the chlorophycean mitochondrial code (translation table 16), alanine in *Pachysolen* nuclear code (translation table 26), and a tripartite serine in the *Candida* nuclear code (translation table 12; (Santos et al., 1999)). These cases, combined with the universal serine disconnection, constitute the complete inventory of amino acid graph disconnections at unit Hamming distance.

3.8.3 Atchley Factor 3 and Serine convergence

Serine has the most extreme Atchley Factor 3 score among the 20 amino acids ($F_3 = -4.760$, 2.24 SD below the mean; (Atchley et al., 2005)), and it is the only amino acid disconnected at $\varepsilon = 1$ under every base-to-bit encoding. This convergence is not coincidental: Factor 3 is a composite of molecular size and codon diversity, so both measures reflect the same underlying anomaly—Serine's disproportionate codon diversity (6 codons in two disconnected families) relative to its small physicochemical footprint. The $\mathrm{GF}(2)^6$ framework provides a structural explanation for why Serine is an outlier on Factor 3, though the two views are complementary rather than independent.

3.9 Retrospective cross-study reanalysis of synthetic genome recoding outcomes

To assess whether the structural properties identified in Sections 3.1–3.5 translate to measurable phenotypic consequences in synthetic biology, we performed a retrospective cross-study reanalysis of eight published genome recoding datasets comprising over 248,000 codon-level observations (Table 8). Each codon substitution was classified by its GF(2)⁶ topology: whether it crosses a connected-component boundary at $\varepsilon = 1$, the change in local physicochemical mismatch cost ($\Delta F_{\mathrm{local}}$, Grantham distance), and the Hamming distance between source and target vectors.

3.9.1 A three-layer decomposition of recoding outcomes

The cross-study reanalysis revealed three separable layers of codon-space structure, each with distinct predictive scope.

3.9.1.1 Boundary crossing: validated but not predictive of transcriptomic perturbation

In the Robertson et al. (2025) Syn57 dataset—the only large-scale dataset with within-study variation in boundary crossing—all 37,146 serine recodings crossed the UCN↔AGY disconnection boundary at $\varepsilon = 1$, while all 22,859 alanine recodings remained within the compact GCN family. This provides

Study	Swap type	n	Boundary?	Key result
Robertson et al. (2025) Syn57	4 Ser + 2 Ala + Stop	60,240	Ser=100%, Ala=0%	Primary contrast
Fredens et al. (2019) Syn61	Ser (TCG/TCA→AGC/AGT)	18,218	All cross	Informative null
Ostrov et al. (2016)	7 codon types	62,214	Mixed	Segment viability
Napolitano et al. (2016)	Arg (AGR→CGN)	12,888	None	SRZ covariates
Grome et al. (2025) Ochre	Stop (TGA→TAA)	1,195	N/A	Growth data
Nyerges et al. (2024)	7 codon types	62,007	Mixed	Multi-omics
Lajoie et al. (2013) C321.ΔA	Stop (UAG→UAA)	321	N/A	Baseline
Frumkin et al. (2018)	Arg (CGU/CGC→CGG)	60	None	Translation efficiency
Ding et al. (2024)	Ser (TCG, mammalian)	Variable	Yes	Cross-organism

Table 8: Published genome recoding datasets analyzed. "Boundary?" indicates whether the synonymous codon swaps cross a connected-component boundary at $\varepsilon = 1$ in GF(2)[6]. †Quantitative extraction pending; included for scope. Only 2 of 8 fully analyzed datasets (Syn57 and Ding) have within-study variation in the boundary-crossing feature.

an exact within-study topology contrast. However, genes containing exclusively serine-type versus exclusively alanine-type recodings did not differ in transcriptomic perturbation (Mann–Whitney $p = 0.40$, $n_{\text{Ser}} = 129$, $n_{\text{Ala}} = 49$), and the fraction of boundary-crossing recodings per gene was uncorrelated with $|\log_2 \text{FC}|$ ($\rho = 0.012$, $p = 0.46$, $n = 3\{,\}510$ genes). In a separate dataset, all 18,218 Syn61 serine recodings (Fredens et al., 2019) crossed the boundary, yet 99.96% were viable (7 corrections needed). Boundary crossing is therefore a coarse geometric property of codon-family organization that is validated as a structural distinction but not as a predictor of transcriptomic perturbation.

3.9.1.2 Local mismatch geometry: a fine-scale burden axis with positive signal

The second layer examines whether reassignment changes the physicochemical quality of the immediate mutational neighborhood. In the Syn57 contrast, serine swaps moved codons into better local Grantham neighborhoods ($\Delta F_{\text{local}} = -37.2 \pm 30.5$), whereas alanine swaps moved codons into worse neighborhoods ($\Delta F_{\text{local}} = +19.0 \pm 0.0$); the distributions were completely separated (Mann–Whitney $U = 0$, $p < 10^{-16}$, $n = 60\{,\}240$). In a topology-fixed setting—arginine synonymous recodings (Napolitano et al., 2016), where all 12,888 AGR→CGN swaps remain within one connected component at $\varepsilon = 1$—local mismatch change correlated with established recoding-burden covariates: ΔF_{local} vs RBS deviation ($\rho = -0.33$, $p < 10^{-15}$) and vs mRNA folding deviation ($\rho = -0.12$, $p < 10^{-15}$). Among the four CGN targets, CGG has the lowest local mismatch cost (246 Grantham units) while CGC has the highest (421), and target-specific RBS deviation spans four orders of magnitude, suggesting that local mismatch and SRZ covariates capture partially independent dimensions of recoding burden.

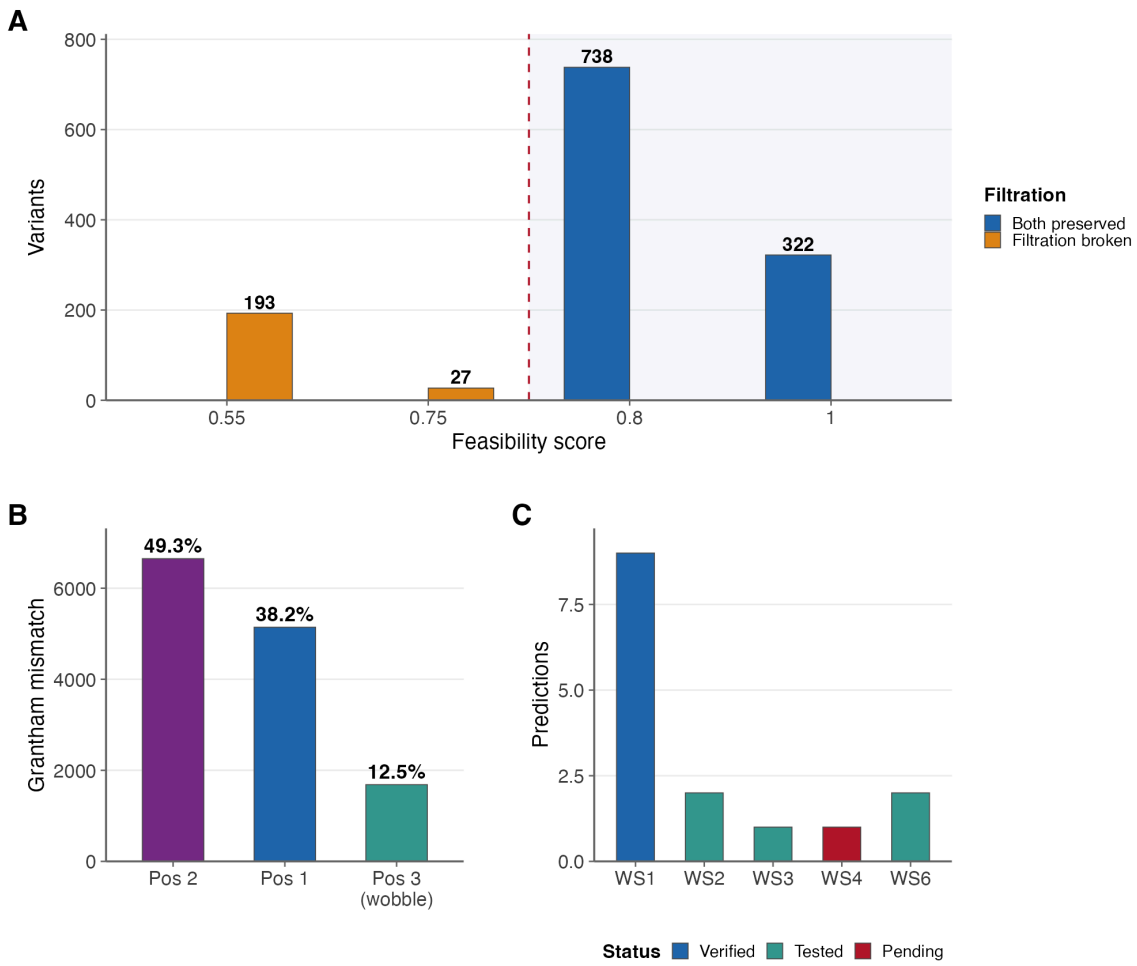

Figure 5: Translational applications and prediction catalogue. **(A)** Feasibility landscape of 1,280 single-codon reassignments from the standard code, colored by degeneracy filtration status. All filtration-preserving variants score ≥ 0.8; all filtration-breaking variants score ≤ 0.75. Dashed line: high-feasibility threshold. **(B)** Grantham mismatch score decomposition by nucleotide position: position 2 dominates (49.3%), consistent with the biochemical hierarchy of mutational impact. **(C)** Prediction catalogue: 15 evaluated claims distributed across 5 workstreams (WS1–WS6), colored by verification status.

3.9.1.3 Design deviations: accessibility-dominated, not topology-directed

To test whether engineered systems escape design constraints along topology-defined low-burden directions, we compared the Syn57 design genome (Data file S2; (Robertson et al., 2025)) against the final verified genome (Data file S8) and identified 727 CDS-level codon differences. After filtering 7 genes with >10 deviations each (structural rearrangements, predominantly *bioF* with 311 changes), 326 genuine point deviations remained (267 synonymous, 59 nonsynonymous). Among the 163 deviations that changed local mismatch, 76 moved to a better neighborhood and 87 to a worse one—no directional bias (two-sided exact binomial $p = 0.43$, Clopper–Pearson 95% CI $[0.39, 0.55]$; Wilcoxon signed-rank $p = 0.67$). This null result was stable across all filtering thresholds tested ($p > 0.5$ at cutoffs 3, 5, 10, 15, 20, and 50 deviations per gene). By contrast, deviations strongly favored mutationally proximate moves: 83% were single-bit changes (Hamming distance 1 under the default encoding; note that Hamming distance is encoding-dependent, though nucleotide edit distance is invariant), and the mean

Hamming distance was 1.21. Thus, realized deviations from the Syn57 design do not preferentially descend toward lower local mismatch; instead, they follow accessibility-dominated escape routes, favoring nearby codon states regardless of neighborhood quality. The Ding et al. (2024) mammalian TCG recoding independently confirms the universality of the serine disconnection across standard-code organisms, providing cross-kingdom validation.

3.9.1.4 Segment-level recoding burden in the Ostrov 57-codon design

As an orthogonal test, we examined segment-level outcomes from the Ostrov et al. (2016) 57-codon genome design. Among 44 segments with fitness data (of 87 total), segments with more recoded codons in essential genes showed a suggestive correlation with worse doubling time ($\rho = 0.34$, raw $p = 0.022$, Bonferroni-corrected $p = 0.066$), while total recoding load showed no association ($\rho = -0.10$, $p = 0.53$). Problem segments (13 with lethal exceptions) trended toward higher essential-gene recoding load (73.5 vs 39.8 recoded sites, Mann–Whitney $p = 0.086$). Among 44 fully characterized segments, 17 showed spontaneous codon reversions, indicating ongoing design instability at positions where the engineered code was under selection pressure.

3.9.2 Summary of cross-study reanalysis scope

The synthetic recoding analyses support a three-layer view: boundary crossing captures a real family-level structural distinction but did not predict transcriptomic perturbation; local mismatch geometry captures a finer-scale burden axis with positive signal; and realized design deviations are dominated by mutational accessibility rather than systematic escape toward lower local mismatch. These layers are partially separable: different structural descriptors appear to matter at different levels of the problem—family topology for the geometry of reassignment space, local mismatch for some aspects of recoding burden, and Hamming accessibility for the short-range routes by which engineered designs deviate or revert.

3.10 Falsified and rejected claims

3.10.1 KRAS–Fano clinical prediction

The conjecture that XOR ("Fano") relationships in $\mathrm{GF}(2)^6$ predict co-mutation enrichment at KRAS G12 sites was tested against 1,670 mutations from MSK-IMPACT (Zehir et al., 2017) and cleanly falsified ($p = 1.0$; details in Supplementary Material). This negative result separates code-level error-minimization (which is real) from mutation-level algebraic predictions (which are not).

3.10.2 Serine distance-4 invariant

The claim that Serine's minimum inter-family Hamming distance (UCN–AGY) equals 4 under all 24 base-to-bit encodings is false. Of the 24 encodings, 16 yield minimum distance 2 and only 8 yield distance 4. The distance-4 result obtains only when both nucleotide pairs distinguishing UCN from AGY ($U \sim A$ and $C \sim G$ in the first two positions) are encoded at maximal Hamming distance. The correct encoding-invariant statement is: Serine is disconnected at $\varepsilon = 1$ under every encoding, and its inter-family distance (≥ 2) is the largest among the three 6-codon amino acids (Leucine and Arginine both have inter-family distance 1).

3.10.3 PSL(2,7) symmetry and holomorphic embedding

The claim that PSL(2,7) is the fundamental symmetry group of the genetic code was pre-rejected by Antoneli and Forger (2011), who showed that PSL(2,7) has no 64-dimensional irreducible representation (its irreps have dimensions 1, 3, 6, 7, 8). The claim that the coordinate-wise map $\mathrm{GF}(2)^6 \to \mathbb{C}^3$ sending base pairs to fourth roots of unity is a holomorphic embedding extending a character of $\mathrm{GF}(8)^*$ is also incorrect: the domain is a finite discrete set (not a complex manifold), and the map fails the character identity $\chi(x + x) = \chi(x)^2$ since $i^2 = -1 \neq 1$.

3.11 Negative results with informative interpretation

The local mismatch cost test—asking whether reassigned codons sit in "worse" Hamming neighborhoods with higher Grantham distance to their neighbors—yields a Mann–Whitney $U = 301$, $p = 0.70$. Reassignment is not driven by local escape from costly neighborhoods. This null result is scientifically informative: it indicates that the topology avoidance constraint (Section 3.4) operates at the global graph-connectivity level, not at the local per-codon level.

4 Discussion

4.1 An information-theoretic view of the genetic code

The central finding of this work is that the standard genetic code minimizes the physicochemical disruption caused by single-bit errors in $\mathrm{GF}(2)^6$ coordinates. This is not a new conclusion—Freeland and Hurst (1998) established error-minimization using nucleotide-level mutation models—but the hypercube representation makes the optimality principle geometrically explicit: the code is a good *coloring* of a structured graph, in the graph-theoretic sense that adjacent vertices (codons differing by one bit) tend to share labels (amino acids) or, when they differ, differ by small physicochemical distances.

The score decomposition (Figure 2) shows that this optimization is concentrated at the second codon position (49.3% of total mismatch) and first position (38.2%), with the wobble position contributing only 12.5%. This gradient mirrors the biochemical hierarchy of mutational impact and is an emergent property of the code's structure rather than a parameter of the model.

The robustness result (Figure 2) demonstrates that optimality is not an artifact of restricting attention to Q_6: when the full K_4^3 mutation graph is considered ($\rho = 1$), the signal strengthens. The code minimizes error not just along the hypercube edges but across the complete space of single-nucleotide substitutions. This error-minimization is complementary to, but distinct from, the finding of Itzkovitz and Alon (2007) that the code is also nearly optimal for carrying parallel regulatory information within protein-coding sequences—a property related to stop-codon identity rather than amino acid physicochemistry.

A natural question is whether the $\mathrm{GF}(2)^6$ framework adds genuine insight beyond what a direct analysis of K_4^3 would provide. We note three advantages. First, the binary decomposition enables the ρ-interpolation that reveals the relationship between Q_6 and K_4^3 optimality as a continuum rather than two disconnected analyses. Second, the Hamming-distance filtration provides a natural persistence parameter (ε) for studying amino acid graph connectivity, yielding the topology avoidance result. Third, the encoding-sensitivity analysis (24 bijections) distinguishes encoding-invariant properties (Serine disconnection) from encoding-dependent ones (distance-4 claim), a distinction invisible in the nucleotide-level representation. However, we emphasize that the Hamming-1/diagonal decomposition

does not correspond to the biological transition/transversion partition: 16 of 24 encodings mix transitions and transversions equally among Hamming-1 edges, while 8 encodings place both transitions on diagonal edges (Supplementary Table S2). The ρ parameter is best interpreted as a diagonal-edge weight, not a transition/transversion weight.

4.2 Evolutionary preservation and topology avoidance

The per-table analysis (Figure 2) shows that 24 of 25 NCBI translation tables maintain coloring optimality under their own block-preserving null, suggesting that codon reassignment events are constrained to preserve error-minimization. The single marginal exception (translation table 3, yeast mitochondrial) is the most extensively reassigned code (6 changes), yet still falls only slightly above the 5% threshold (quantile 7.4%).

The topology avoidance result (Figure 3) provides a mechanistic complement: natural reassignments avoid creating new amino acid disconnections at a rate far below chance expectation. Importantly, this depletion is not confined to the Q_6 representation: under the full K_4^3 single-nucleotide mutation graph, the observed proportion of topology-breaking reassignments remains 22.2% against a possible rate of 66.1% ($\mathrm{RR} = 0.34$, 95% CI $[0.17, 0.68]$, $p \leq 10^{-4}$). The signal is somewhat attenuated relative to the Q_6 result ($\mathrm{RR} = 0.31$) because some Q_6-disconnected pairs become connected when all single-nucleotide edges are admitted, but the main qualitative conclusion survives: the connected-component structure of amino acid codon families is functionally important, regardless of whether connectivity is defined on the hypercube subgraph or on the biologically fuller single-substitution graph. The yeast mitochondrial threonine reassignment illustrates this cost: the CUN→Thr change required acquisition of a novel $\mathrm{tRNA}^{\mathrm{Thr}}$ derived from $\mathrm{tRNA}^{\mathrm{His}}$ via anticodon mutation (Su et al., 2011), creating the topology-breaking disconnection that makes translation table 3 the sole marginal outlier in the per-table optimality analysis.

Together, these analyses support a four-part picture: the standard code is unusually low-cost, most variant codes preserve that structure, realized natural reassignments avoid topology-disrupting moves, and event-level explanatory modeling shows that this avoidance carries independent information beyond physicochemical optimization. The conditional logit analysis (Table 5) makes the last step precise: the topology term retains major explanatory value ($\Delta\mathrm{AICc} = 100$ relative to a physicochemistry-only model) even after accounting for local physicochemical cost, while physicochemical cost also retains value ($\Delta\mathrm{AICc} = 67$) after accounting for topology. The two signals are therefore complementary rather than redundant, consistent with the weak correlation ($r_s = 0.15$) between the feature classes across the candidate landscape. This does not by itself prove direct selection on topology as an abstract graph property; topology may still proxy aspects of decoding architecture—such as tRNA cross-recognition range or ribosomal A-site constraints—not explicitly modeled here. However, within the tested event landscape, topology avoidance cannot be dismissed as a by-product of physicochemical optimization. Topology avoidance is not a geometric restatement of error minimization, but a complementary constraint on how genetic codes can change while remaining evolutionarily accessible.

4.3 Mechanistic implications: tRNA compensation

The tRNA enrichment result (Figure 3, panel D) links the geometric observation to molecular mechanism. In several variant-code lineages where codon reassignment disrupts connectivity, expanded tRNA gene repertoires for the affected amino acid are observed, consistent with compensatory gene duplication as one evolutionary route to accommodation. The extreme case of *Tetrahymena*

thermophila (54 Gln tRNAs, including 39 suppressors) illustrates the scale of genomic response required to service a split codon family.

However, the boundary cases—*Blastocrithidia nonstop* (anticodon stem shortening; (Kachale et al., 2023)) and *Mycoplasma* species (anticodon modification)—show that tRNA gene duplication is not the only evolutionary solution. The three-tier pattern (duplication in large genomes, structural modification in intermediate genomes, base modification in minimal genomes) suggests that genome size constrains the available mechanistic repertoire for codon reassignment. This pattern is now supported by 18 tRNAscan-SE–verified genomes across 5 variant genetic codes (Tables 4, 6, 10, 15, and 31), including 6 *Euplotes* species (UGA→Cys) that each carry 1–4 TCA-anticodon tRNA-Cys genes dedicated to reading UGA.

4.4 Synthetic recoding outcomes: a three-layer interpretation

The retrospective cross-study reanalysis helps delimit which structural quantities in GF(2)^6 are functionally relevant and at which biological layer. Boundary crossing is a valid codon-space distinction—serine and alanine provide a clean contrast in which one class always crosses a family boundary and the other never does—yet this distinction did not predict RNA-seq perturbation in Syn57. We therefore do not interpret codon-family boundary crossing as a general transcriptomic burden variable. Its relevance, if any, likely lies at a different biological layer, such as decoding architecture, translational fidelity, or evolutionary accessibility, rather than steady-state expression change.

By contrast, local mismatch geometry showed a clearer positive signal. In both the Syn57 contrast and the topology-fixed arginine setting, codon changes differed systematically in the quality of the physicochemical neighborhood they entered, and these differences aligned with established recoding-burden covariates (Napolitano et al., 2016). This suggests that local mismatch is best understood as a fine-scale recoding-friction variable: not a universal master predictor, but a meaningful descriptor of neighborhood quality within an already specified codon-family structure.

The design-deviation analysis further clarifies scope. Genuine deviations from the Syn57 design were overwhelmingly short-range (83% single-bit in GF(2)^6), with no directional bias in mismatch change. This indicates that realized escape is governed more by mutational accessibility than by systematic optimization of local neighborhood cost. In other words, once a design is under strain, the available exits appear to be chosen primarily by mutational proximity, not by a global preference for lower mismatch.

These findings argue against treating "topology" as a single explanatory variable. Instead, different structural descriptors appear to matter at different levels: family topology for the geometry of reassignment space (Section 3.4), local mismatch for some aspects of recoding burden (Section 3.7.2), and Hamming accessibility for the short-range routes by which engineered designs deviate. The fact that the Fredens et al. (2019) Syn61 tolerated 18,218 boundary-crossing serine swaps with 99.96% success, while the NCBI variant codes show 3.3-fold depletion of topology-breaking changes over evolutionary timescales, suggests that boundary crossing may be more relevant to evolutionary trajectory constraints than to acute engineering costs.

4.5 Scope and falsified claims

Six of 15 evaluated claims were rejected, falsified, or tautological (shaded rows in Table 1; full details in Supplementary Material). These failures help delimit the framework's explanatory scope. The KRAS–Fano prediction ($p = 1.0$) cleanly separates code-level error-minimization from mutation-level algebraic predictions. The Serine distance-4 invariant, falsified by 16 of 24 encodings giving distance

2, illustrates the importance of systematic encoding-sensitivity testing. The PSL(2,7) and holomorphic embedding claims were pre-rejected by existing mathematical results (Antoneli and Forger, 2011).

4.6 Limitations

Several limitations should be noted. First, the choice of base-to-bit encoding is not unique, and while the coloring optimality result holds across all 24 encodings, the specific score values and rank orderings are encoding-dependent (full encoding-sensitivity results across all 24 bijections are provided in the Supplementary Material). Second, the block-preserving null model, while standard in the field (Freeland and Hurst, 1998), preserves more structure than a fully random code and may understate the degree of optimality. Third, the tRNA enrichment result, while robust to pairing selection (worst-case MIS $p = 0.045$), relies on a small number of independent pairings ($n = 6$) with limited statistical power; this result is appropriately classified as suggestive and should not be interpreted as demonstrating a universal compensation mechanism. Fourth, the conditional logit model (Section 3.5) is an event-level explanatory model, not direct proof of biological causality; the candidate universe comprises all $\approx 1\{,\}280$ single-codon reassignments regardless of mechanistic plausibility, and the tRNA-complexity proxy (Hamming distance to nearest target-AA codon) is heuristic rather than mechanistically grounded. The finding that this particular proxy did not improve model fit does not exclude the possibility that a richer model of tRNA repertoire change would contribute explanatory value.

Whether the code's optimality reflects adaptive selection or non-adaptive carry-over from a primordially constrained starting point (Koonin and Novozhilov, 2009) cannot be resolved by cross-sectional data alone. Multiple theoretical frameworks address this question: Sella and Ardell (2006) showed that coevolution of genes and codes generates error-correcting structure resembling the standard code; Vetsigian et al. (2006) argued that communal evolution via horizontal gene transfer accounts for both universality and optimality; and Novozhilov et al. (2007) found the standard code sits roughly halfway up a local peak in a rugged fitness landscape (see (Di~Giulio, 2005) for a review of origin theories). The topology avoidance result is consistent with all these frameworks: it demonstrates a constraint on reassignment trajectories but does not distinguish whether the constraint is selective or structural. Novozhilov and Koonin (2009) showed that putative primordial 16-supercodon codes are "nearly optimal" even without direct selection, suggesting the current code's optimality may be partly inherited.

The $\mathrm{GF}(2)^6$ representation is best understood as an analytical decomposition tool rather than a claim about the biological primacy of binary coordinates. It enables the ρ-sweep interpolation between Q_6 and K_4^3 and facilitates systematic encoding-sensitivity analysis. However, we caution that the Hamming-1/diagonal decomposition in $\mathrm{GF}(2)^6$ does not correspond to the biological transition/transversion partition: 16 of 24 encodings mix transitions and transversions equally among Hamming-1 edges, while 8 encodings place both transitions on diagonal edges. The ρ parameter should therefore be interpreted as a diagonal-edge weight, not a proxy for transition/transversion bias. The underlying biology is the assignment of chemically similar amino acids to mutationally proximate codons—a property that holds regardless of the coordinate system, as confirmed by the K4^3 topology avoidance result.

5 Conclusion

Across four amino acid distance metrics and across a continuum of mutation graphs spanning the hypercube and the complete single-nucleotide substitution graph, the standard genetic code consis-

tently falls in the extreme low-cost tail of block-preserving null models ($p \leq 0.006$ under all metrics and all ρ values). That near-optimality is largely preserved across naturally variant translation tables (24 of 25 significant after FDR correction), implying that most reassignment histories remain within a constrained region of code space rather than freely exploring it.

Two constraints on reassignment trajectories emerge from the event-level analyses. First, evolution favors physicochemical smoothness: reassigned codons tend to enter neighborhoods with lower local mismatch cost. Second, evolution avoids fragmenting amino acid codon families: topology-breaking moves are ≈3-fold depleted relative to the landscape of possibilities, and discrete-choice modeling shows that this topology term improves explanatory fit even after accounting for physicochemical cost ($\Delta\mathrm{AICc} \geq 67$), and vice versa. This supports the view that topology avoidance is not a geometric restatement of physicochemical optimization, while remaining agnostic about the precise mechanism —whether tRNA cross-recognition range, ribosomal decoding constraints, or other aspects of translational architecture not explicitly modeled here. Several variant-code lineages with topology-breaking reassignments show expanded tRNA gene repertoires for the reassigned amino acid (worst-case MIS Stouffer $p = 0.045$), consistent with compensatory accommodation routes.

In engineered recoding datasets (>248,000 codon-level observations across eight studies), these same codon-space descriptors appear to operate at different biological layers: family boundary crossing is a real structural distinction but was not predictive of transcriptomic perturbation, whereas local neighborhood mismatch more often aligned with established recoding-burden covariates and design deviations were dominated by mutational accessibility. Natural code evolution appears constrained along two partly independent axes: physicochemical smoothness and codon-family topological integrity.

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Data availability

All code and data are available in the `codon-topo` repository (version 0.3.1). Install with `pip install -e ".[all]"` or `uv sync --all-extras`. Analyses are fully reproducible via `codon-topo all --output-dir=./output --seed=135325`, which generates all JSON data files including `manuscript_stats.json` from which every inline statistic in this manuscript is rendered. NCBI genome assembly accessions are listed in Table 7.

Declaration of competing interest

The authors declare no competing interests.

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